

BMO2 2020 Problem 1

Problem: A sequence $a_1, a_2, a_3, \dots$ has $a_1 > 2$ and satisfies

$$a_{n+1} = \frac{a_n(a_n - 1)}{2}$$

for all positive integers n . For which values of a_1 are all the terms of the sequence odd integers?

Solution:

Proof. Suppose all terms are odd integers. Then $a_n + 2$ is odd for every n .

Now,

$$\begin{aligned} a_{n+1} - 3 &= \frac{a_n(a_n - 1)}{2} - 3 \\ &= \frac{a_n^2 - a_n - 6}{2} \\ &= \frac{(a_n - 3)(a_n + 2)}{2}. \end{aligned}$$

Since $a_n + 2$ is odd, passing from $a_n - 3$ to $a_{n+1} - 3$ removes exactly one factor of 2.

Thus, if $a_1 - 3$ were nonzero, then after enough steps all of its factors of 2 would be removed. But every $a_n - 3$ must still be an integer. By infinite descent, this is impossible unless

$$a_1 - 3 = 0.$$

Hence

$$a_1 = 3.$$

Checking, if $a_1 = 3$, then

$$a_2 = \frac{3(3-1)}{2} = 3,$$

so the sequence is constantly 3.

Therefore,

$$\boxed{a_1 = 3}.$$

□